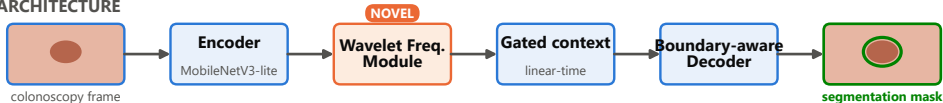


WaveLite-DG

A Lightweight Wavelet-Guided Hybrid Network for Generalizable, Real-Time Polyp Segmentation

ARCHITECTURE



CLINICAL CHALLENGE

TRAIN · 2 source centers



↓ *domain shift*

DEPLOY · 3 unseen centers



Scanner & population shift degrade accuracy at unseen hospitals.

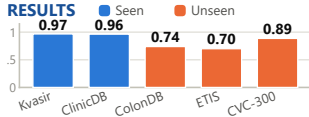
WAVELET FREQUENCY MODULE



Style lives in low frequencies; anatomy in high — perturb one, preserve the other → domain-invariant.

+ boundary-aware decoder & gated context block global context at low parameter cost.

RESULTS



7.65 M
parameters

1.93 G
FLOPs (lowest)

2 → 3
train → unseen